MATH 455 Statistical Learning Final Project

Overview

This project is an opportunity to apply statistical learning techniques to a dataset of your choice. Working in teams, you will pose two or more interesting questions about the data and use methods from the course to investigate them.

Your analysis should include appropriate data exploration and visualization, followed by one or more statistical learning methods such as regression, classification, clustering, dimension reduction, tree-based methods, or other methods appropriate to your questions. You should compare reasonable approaches when appropriate and explain the choices you make.

The project should represent original work by your group. This does not require developing a new statistical method. Originality may come from the questions you pose, the dataset you analyze, the combination of methods you use, and the conclusions you draw.

Schedule

- **Project proposal due:** Tuesday, November 3rd
- **Presentations:** Tuesday, December 1 and Thursday, December 3
- **Final write-up due:** Friday, December 4 at 11:59 PM.

Proposal

Submit a short proposal, no more than one page, containing:

- group member names (**either 3 or 4 members are permitted**);
- the dataset you plan to use and where it comes from;
- the questions you want to investigate;
- the response or outcome of interest, if applicable;
- the statistical learning methods you expect to consider; and
- a brief description of how you plan to evaluate or compare those methods.

The proposal is intended to make sure the project is feasible and sufficiently substantive. Your final analysis may change as you learn more about the data.

Written Report

Write a paper **of length no more than 10 pages**, including figures and citations, describing your data, questions, analysis, and findings. Explain and justify important modeling choices, compare methods where appropriate, discuss model performance and parameter selection, and identify important limitations. Your conclusions should be supported by the analysis rather than by model output alone. Beyond these requirements, you may organize the report in whatever way best supports a clear and coherent presentation of the analysis. All group members must participate in the writing.

Presentation

Give a presentation of approximately **15 minutes, followed by five minutes of questions.** All group members must participate in the presentation.

Your presentation should make the substantive questions, basic approach, main findings, and important limitations understandable without assuming familiarity with the technical methods used in the paper. Specialized terminology should be explained when needed, and figures or examples should be used where they communicate the ideas more clearly than mathematical or computational details.

Submission

Submit the final report as a PDF. You should also submit the code or notebook used to produce the main analysis as a `.zip`.

Each group member should be able to explain the main modeling choices, results, and limitations of the project.

Individual assessment.

Although the project is completed in groups, each student will be assessed individually as well as collectively. Every group member must participate substantively in the final presentation and should be prepared to answer questions about the group's data, methods, results, and conclusions. The final written report must also include a **contribution and attribution statement** briefly describing each student's contributions and identifying any substantial use of generative AI or other external tools. Individual project grades may differ within a group based on each student's presentation, responses to questions, contribution statement, and demonstrated understanding of the project as a whole. Each student's presentation performance and demonstrated understanding will be assessed individually. This, combined with the contribution statement may be used to adjust individual grades when contributions differ substantially.

Use of Generative AI

You may use generative AI tools (such as ChatGPT, Claude, or GitHub Copilot) to assist with your project. Appropriate uses include brainstorming ideas, clarifying statistical concepts, writing or debugging code, suggesting analytical approaches, and improving the clarity of your writing.

However, the substantive intellectual work of the project must come from your group. This includes developing meaningful research questions, making and justifying modeling decisions, evaluating results, and drawing conclusions. AI should support these activities rather than replace them.

You are responsible for the accuracy and validity of everything you submit, regardless of whether AI was used to produce it. In particular, you should carefully review and understand any AI-generated code, verify statistical results, and critically evaluate AI-generated suggestions or interpretations.

Your final report must include a brief statement describing any substantial use of generative AI, including the tools used and how they contributed to the project. You do not need to provide a complete record of your interactions with AI.

Every group member is expected to understand the project as a whole and be able to explain and defend its main analytical decisions, methods, and conclusions. This includes material developed with AI assistance. Your demonstrated understanding during the presentation and questions will be an important component of your individual assessment.